Ἀπολλώνιος ὁ Περγαῖος

Apollonius of Perga and the Conics

eul_aid: kle

Period: *Ἑλληνιστική* Hellenistic
Dialect: *Κοινὴ τεχνική* Technical Koine
Domain: *Μαθηματικά* Mathematics
Format: *Πραγματεία* Treatise

Apollonius of Perga was a Greek mathematician and astronomer active in the late 3rd and early 2nd centuries BCE. He was born in Perga, in modern-day Turkey, and worked primarily in Alexandria, the intellectual center of the Hellenistic world. He studied under the successors of Euclid and also spent time in Pergamum, where he enjoyed the patronage of its king.

His fame rests on his work in geometry, particularly his systematic study of conic sections—the curves known as ellipses, parabolas, and hyperbolas. His major and only substantially surviving work is the Conics, an eight-book treatise. The first four books survive in Greek, while three more survive through an Arabic translation; the final book is lost. Apollonius is credited with introducing the names "ellipse," "parabola," and "hyperbola," terms still used today.

He also wrote several other advanced geometrical works, such as On the Cutting-off of a Ratio and Tangencies, but most survive only in fragments, Arabic translations, or are known only by title. According to modern scholars, Apollonius is considered one of the greatest mathematicians of antiquity.

His Conics organized and greatly expanded the theory of conic sections, treating them as plane curves with defined properties. This work became the standard reference on the subject for nearly two thousand years, deeply influencing later Greek, Islamic, and Renaissance science. It has been interpreted as a foundational text for astronomy, notably used by Johannes Kepler in his laws of planetary motion.

Works (1)